

travel booking

Req2Design · IEEE 830-1998 aligned

Document Control

Version	Date	Author	Change Summary
1.0	2026-04-09	usman asif	Initial generated draft for structured review.

Document ID: SRS-20260409-162109

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INTRODUCTION

Purpose: To define the requirements for a travel booking system that allows users to search, book, and manage their trips.

Scope: This document covers the functional and non-functional requirements for the travel booking system.

Definitions/Acronyms: API

- Application Programming Interface; CRM
- Customer Relationship Management

References: IEEE Standard for Software Requirements Specifications, Version 1.1, IEEE Std 830-1998

Overview: The travel booking system aims to provide users with a seamless experience for planning and managing their journeys.

OVERALL DESCRIPTION

Product Perspective: The product is a web-based travel booking system designed for individual travelers who need assistance in finding and purchasing flights, accommodations, and related services.

Product Functions: The system provides functionalities for searching multi-city itineraries, comparing fares, reserving seats, adding insurance, receiving e-tickets, handling voucher redemptions, fare changes, offering alternative options during checkout, sending trip reminders, allowing simple changes/cancellations, storing passenger profiles, and displaying seat maps.

User Characteristics: Target users are individuals aged 18 and above, interested in traveling both domestically and internationally. They have varying levels of technical proficiency but prefer ease-of-use over advanced customization.

Constraints: The system must operate on standard PCs with minimum hardware specifications and support common web browsers and operating systems. Communication with external services occurs via secure protocols.

Assumptions/Dependencies: Users have access to the Internet and possess necessary financial means to complete transactions. Third-party systems providing flight data and other services are reliable and up-to-date.

SPECIFIC REQUIREMENTS

External Interface Requirements

- **User Interface:** Modern aesthetic, responsive design compatible with Windows, macOS, and Linux.
- **Hardware Interface:** Minimum configuration includes 4GB RAM and Intel i5 processor; stable internet connection required.
- **Software Interface:** Integration with airline databases, payment gateways, and CRM tools; supports common web browsers and operating systems.
- **Communication Interface:** HTTP/HTTPS, JSON/XML data formats, WebSocket for real-time updates, SSL/TLS for secure transactions.

Functional Requirements

FR-01 Search Multi-City Itinerary

- Users enter origin, destination, dates, and number of passengers.
- System retrieves available flights from multiple airlines.
- Display results sorted by price, duration, and layovers.

Priority: High

FR-02 Compare Fares

- Users select preferred itinerary and view detailed fare comparisons.
- Include baggage allowance, refund policy, and additional fees.
- Highlight differences between selected options.

Priority: Medium

FR-03 Reserve Seats

- After selecting a seat map, users confirm their choice.
- System reserves seats based on availability.
- Notify user if reservation fails due to lack of space.

Priority: High

FR-04 Add Insurance

- Users choose optional travel insurance.
- Calculate premium based on trip details.
- Confirm purchase and update total cost.

Priority: Low

FR-05 Receive E-Ticket

- Upon successful transaction, generate and send electronic ticket.
- Store ticket information securely for future reference.
- Provide option to download or print tickets.

Priority: Medium

FR-06 Voucher Redemption

- Users apply valid vouchers towards their booking.
- Validate voucher against current pricing and inventory.
- Update total cost after applying discount.

Priority: Low

FR-07 Fare Change Handling

- Handle requests for fare adjustments post-purchase.
- Ensure new prices reflect current market rates.
- Communicate changes to affected customers promptly.

Priority: Medium

FR-08 Alternative Options During Checkout

- Offer alternatives if chosen flights become unavailable.
- Present substitute options within the same time frame.
- Allow users to make selections without losing progress.

Priority: Medium

FR-09 Trip Reminders

- Send automated notifications before departure.
- Customize messages according to user preferences.

- Enable rescheduling of reminder times.

Priority: Low

FR-10 Simple Changes/Cancellations

- Allow users to modify basic booking details.
- Process cancellations efficiently while maintaining records.
- Refund amounts calculated accurately based on policy.

Priority: Medium

FR-11 Passenger Profile Storage

- Collect and store personal information for repeat customers.
- Use profile data to personalize future searches and offers.
- Protect sensitive data with encryption.

Priority: High

FR-12 Seat Map Selection

- Display available seats visually.
- Allow selection of specific rows and columns.
- Show status of each seat (available, reserved, blocked).

Priority: High

Non-Functional Requirements

Usability:

- Intuitive navigation and layout.
- Clear error messages and prompts.

Reliability:

- Minimize downtime; aim for 99% uptime.
- Robust validation checks to prevent errors.

Performance:

- Load pages within 5 seconds under normal conditions.

- Complete complex operations within 30 seconds.

Portability:

- Compatible with Windows, macOS, and Linux.
- Responsive design supporting mobile and desktop views.

SYSTEM FEATURES

Feature 1: Search Multi-City Itinerary

- Allows users to enter origin, destination, dates, and number of passengers.
- Retrieves available flights from multiple airlines.
- Displays results sorted by price, duration, and layovers.
- Ensures intuitive navigation and layout.
- Provides robust validation checks to prevent errors.

Feature 2: Compare Fares

- Enables users to select preferred itinerary and view detailed fare comparisons.
- Includes baggage allowance, refund policy, and additional fees.
- Highlights differences between selected options.
- Offers clear error messages and prompts.

Feature 3: Reserve Seats

- Facilitates confirmation of seat choices made by users.
- Reserves seats based on availability.
- Notifies users if reservations fail due to lack of space.
- Maintains efficient processing and record keeping.

Feature 4: Add Insurance

- Supports optional travel insurance purchases.
- Calculates premiums based on trip details.
- Updates total costs upon confirmation.
- Ensures accurate calculations and prompt responses.

Feature 5: Receive E-Ticket

- Generates and sends electronic tickets upon successful transactions.
- Stores ticket information securely for future reference.
- Provides options to download or print tickets.
- Guarantees secure storage and easy accessibility.

Feature 6: Voucher Redemption

- Allows users to apply valid vouchers towards bookings.
- Validates vouchers against current pricing and inventory.
- Adjusts total costs accordingly.
- Ensures timely and accurate processing.

Feature 7: Fare Change Handling

- Manages fare adjustment requests post-purchase.
- Reflects updated prices reflecting current market rates.
- Communicates changes effectively to affected customers.
- Delivers immediate feedback and support.

Feature 8: Alternative Options During Checkout

- Presents substitute options when original choices become unavailable.
- Keeps users informed about ongoing processes.
- Preserves user progress throughout the booking journey.
- Ensures smooth transitions and minimal disruptions.

Feature 9: Trip Reminders

- Sends automated notifications before departures.
- Customizes messages according to user preferences.
- Allows rescheduling of reminder times.
- Enhances user convenience and satisfaction.

Feature 10: Simple Changes/Cancellations

- Enables modifications to basic booking details.
- Processes cancellations quickly and accurately.
- Calculates refunds correctly based on policies.
- Improves overall efficiency and user trust.

Feature 11: Passenger Profile Storage

- Collects and stores personal information for repeat customers.
- Utilizes stored data to personalize future searches and offers.

- Encrypts sensitive data for protection.
- Safeguards privacy and maintains confidentiality.

Feature 12: Seat Map Selection

- Displays visual representations of available seats.
- Allows selection of specific rows and columns.
- Shows statuses of each seat (available, reserved, blocked).
- Ensures clarity and ease of use for all users.

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